DEEP LEARNING FOR COMPUTER VISION

Uncertainty-Aware Object Detection

Combining Aleatoric & Epistemic Uncertainty in Deformable DETR

7 Members

Initial Project Presentation

Team Members

Deep Learning for Computer Vision Course Project

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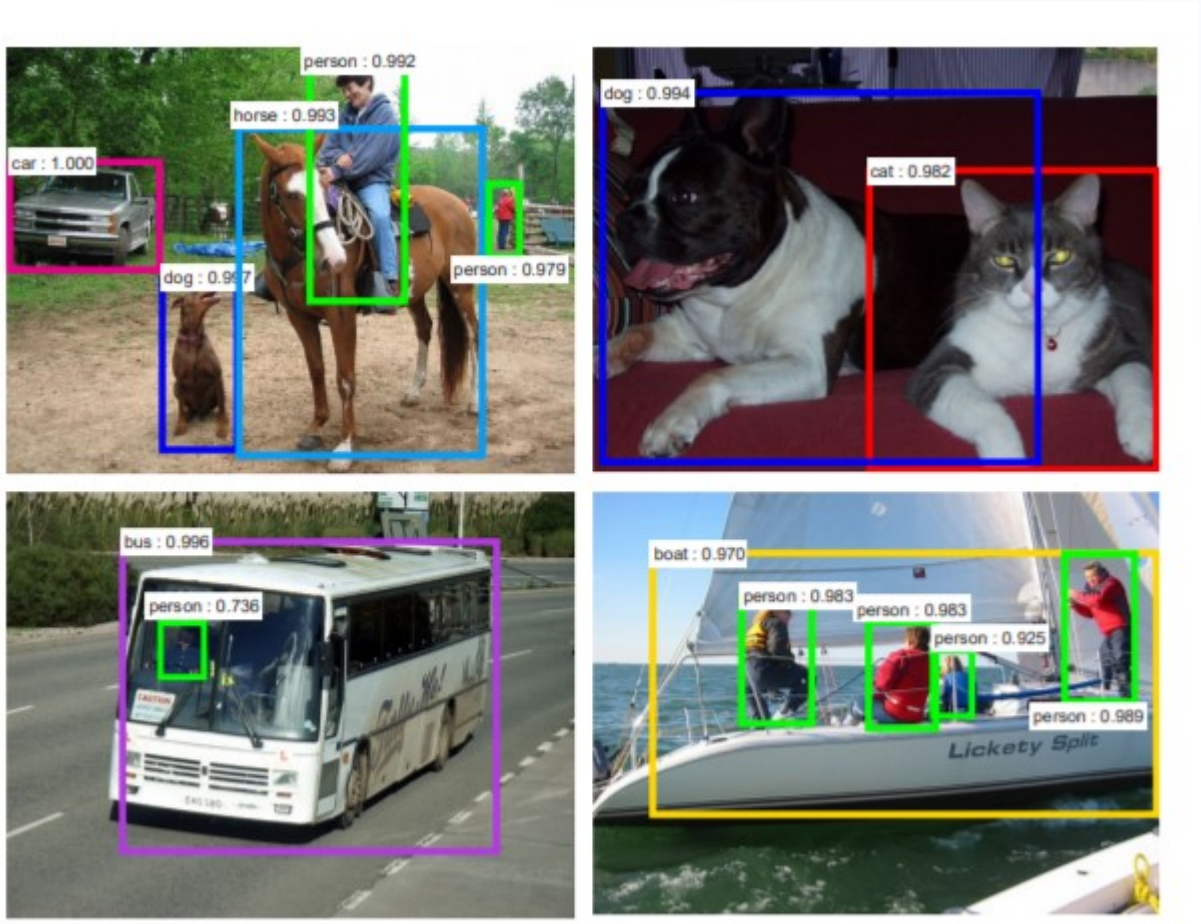
Problem Statement

Safety-Critical Failures

Build an object detector that knows when it should NOT be trusted by separating image-level noise from model-level ignorance, providing calibrated confidence scores.

- **Overconfident Scores:** Current detectors output a softmax score (e.g. 80%) that doesn't reflect true reliability. They can be wrong with no warning.
- **No Uncertainty Separation:** A blurry image (data problem) and a rare object class (model problem) require different responses, but are treated identically.
- **Poor Calibration (High ECE):** For DETR and Deformable DETR, '80% confident' does not mean 80% accurate. They are systematically overconfident.
- **Unsafe for Deployment:** In autonomous driving or medical imaging, silent failures are dangerous.



Research Paper / Author	Core Methodology	Key Findings	Primary Limitations
Retinexformer: One-Stage Retinex-Based Transformer for Low-Light Image Enhancement	Uses Retinex theory with a transformer to separate illumination and reflectance for low-light enhancement.	Improves visibility and feature quality in low-light images, helping downstream vision tasks.	Designed for image enhancement, not object detection; adds preprocessing overhead.
U-Net: Convolutional Networks for Biomedical Image Segmentation	Encoder–decoder CNN with skip connections for precise image segmentation.	Performs well even with limited training data and became a foundational segmentation architecture.	Designed for segmentation rather than detection; does not model prediction uncertainty
GenConViT: Generative Convolutional Vision Transformer for Deepfake Detection	Hybrid model combining CNN and Vision Transformer to detect deepfake artifacts in images and videos.	Achieves strong accuracy in detecting manipulated media using spatial and generative features.	Focused on classification tasks; computationally heavy and not designed for object detection
End-to-End Object Detection with Transformers (DETR)	Transformer encoder-decoder architecture for object detection using object queries and bipartite matching loss	Removes anchors and NMS, enabling simple end-to-end detection framework.	Very slow training (~500 epochs) and weak performance on small objects.
Deformable DETR: Deformable Transformers for End-to-End Object Detection	Introduces deformable attention where each query attends to a small set of reference points instead of full attention.	Faster convergence (~50 epochs) and improved small-object detection compared to DETR.	Still computationally expensive and does not estimate prediction uncertainty.

Research Paper / Author	Core Methodology	Key Findings	Primary Limitations
What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?	Bayesian framework modeling aleatoric uncertainty via variance prediction and epistemic uncertainty using Monte Carlo dropoout	Demonstrates improved robustness and uncertainty estimation in vision tasks like segmentation and depth estimation.	Not designed for object detection models; Monte Carlo sampling increases inference cost.
Bayes R-CNN: Uncertainty-Aware Bayesian Object Detection for Remote Sensing	Extends Faster R-CNN with Bayesian layers and multi-resolution feature extraction for uncertainty estimation.	Improved detection accuracy and interpretability in remote sensing datasets.	Very complex architecture; computationally heavy and tested only on specialized remote-sensing datasets.
Gaussian YOLOv3: Localization Uncertainty for Object Detection	Models bounding box prediction as a Gaussian distribution and predicts variance along with bounding box coordinates.	Improved mAP by ~3–3.5 points on KITTI and BDD datasets while running in real time (~42 FPS).	Only models localization (aleatoric) uncertainty; does not capture epistemic uncertainty; assumes Gaussian noise for all objects.
An Uncertainty-Weighted Decision Transformer for Navigation in Dense, Complex Driving Scenarios	Decision Transformer using entropy-based uncertainty weighting during training.	Improves reward, stability, and reduces collision rate in dense traffic navigation tasks.	Tested mainly in simulation; requires large datasets and high computation.
Capturing Object Detection Uncertainty in Multi-Layer Grid Maps	Uses CNNs to estimate classification, pose, and shape uncertainty in grid-map representations used in autonomous driving.	Shows that uncertainty helps improve robustness and trajectory planning safety.	Designed specifically for grid-map sensor fusion data; not applicable directly to image-based object detectors.

Research Paper / Author	Core Methodology	Key Findings	Primary Limitations
Towards Safe Autonomous Driving: Capture Uncertainty in Deep Neural Networks for LiDAR 3D Vehicle Detection	Bayesian neural network applied to LiDAR 3D object detection with aleatoric and epistemic uncertainty modeling.	Modeling aleatoric uncertainty improved detection accuracy by ~1–5%.	Focused on LiDAR data and 3D detection; not directly applicable to 2D image detectors.
Uncertainty-Aware Gradient Stabilization — Huixin Sun et al., ICCV 2025	Reformulates localization as discrete intervals; Uncertainty-Minimization loss and uncertainty-guided refinement stabilize gradients for tiny boxes	Reduces gradient variance ≈2.9×; improves small-object AP (≈+2.6 on VisDrone); generalizes across detectors.	Adds UM & UR modules (more complexity); discretization may slightly reduce fine localization precision; small-object focus.
Calibrated Detection Transformer (Cal-DETR) — Munir et al., NeurIPS 2023	Train-time calibration: uncertainty quantification, uncertainty-guided logit modulation, plus logit-mixing regularizer for DETR variants.	Significantly lowers calibration error (ECE); maintains AP; improves in- and out-of-domain confidence reliability.	Adds training overhead and extra losses; needs tuning of modulation/ mixing hyperparameters for best results.
GroupEnsemble: — Yang et al., arXiv 2026.	Run multiple object-query groups in parallel through shared decoder; attention masks keep groups independent producing ensemble outputs.	Produces ensemble - quality uncertainty in one forward pass; better-cost-accuracy tradeoff than naive ensembles or MC-dropout.	Increases memory and inference overhead; assumes transformer parallelism; slightly more complex integration than single models.
Out-of-Distribution Object Detection through Bayesian Uncertainty Estimation — Tianhao Zhang et al.,	Samples detector weights from Gaussian distributions to estimate Bayesian uncertainty and detect out-of-distribution objects.	Improves OOD detection performance; reduces FPR95 and increases AUROC when testing detectors on unseen datasets.	Requires weight sampling during inference; additional computation compared to standard deterministic detectors.

Research Paper / Author	Core Methodology	Key Findings	Primary Limitations
E-DETR: Evidential Deep Learning for End-to-End Uncertainty Estimation in Object Detection	Extends DETR with evidential deep learning to estimate uncertainty distributions instead of point predictions. Uses Beta distribution for classification and NIG distribution for bounding box regression.	Provides end-to-end uncertainty estimation in object detection. Improves calibration, robustness, and generalization, especially on imbalanced datasets.	Sensitive to class imbalance and requires careful tuning of evidential regularization. Additional uncertainty modeling increases training complexity.
Uncertainty Meets Diversity: A Comprehensive Active Learning Framework for Indoor 3D Object Detection	Active learning framework combining uncertainty-based sampling with diversity selection using a class-aware adaptive prototype bank.	Improves label efficiency and detection performance by selecting informative and diverse indoor 3D scenes.	Requires iterative retraining and annotation cycles; effectiveness depends on uncertainty estimation quality.
Monte Carlo DropBlock for Modelling Uncertainty in Object Detection – Deepshikha Kumari et al.	Introduces MC-DropBlock, applying DropBlock regularization during training and inference to perform multiple stochastic forward passes and estimate uncertainty in object detection models.	Improves uncertainty estimation and calibration in CNN-based detectors. Enhances robustness and helps detect unreliable predictions and out-of-distribution inputs.	Requires multiple inference passes which increases computational cost. Performance depends on correct tuning of DropBlock parameters
Exploring Aleatoric Uncertainty in Object Detection via Vision Foundation Models	Uses foundation-model features and Mahalanobis distance to estimate aleatoric data uncertainty for each object instance.	Enables uncertainty-aware sample filtering and adaptive training; improves robustness against noisy annotations and difficult objects.	Depends on quality of foundation-model features; additional feature extraction and clustering increase computation.

Research Paper / Author	Core Methodology	Key Findings	Primary Limitations
DETR with Improved DeNoising Anchor Boxes for End-to-End Object Detection — Hao Zhang et al., ICLR 2023	Improves DETR by introducing contrastive denoising training, mixed query selection for anchor initialization, and improved bounding-box prediction, enabling faster convergence and stronger detection performance.	Achieves state-of-the-art object detection accuracy (e.g., >63 AP on COCO with large models) while training significantly faster than earlier DETR variants.	Still computationally heavy; requires large datasets and strong backbones (e.g., Swin Transformer) to reach top performance.
YOLO: You Only Look Once (Joseph Redmon et al.)	Single CNN predicts bounding boxes and classes from whole image using an S×S grid.	Real-time detection, simple end-to-end pipeline, fewer background errors.	Poor localization for small objects, grid limits multiple nearby detections.

Comparison of Existing Approaches

The Gap Our Project Fills

No transformer detector has been extended with both aleatoric AND epistemic uncertainty estimation.

- ✗ DETR and Deformable DETR lack calibration completely.
- ✗ Kendall & Gal's foundational method was never applied to object detection.
- ✓ MC Dropout at T=50 is too slow. We show T=10 decoder-only dropout achieves comparable quality at 10x overhead.

Understanding The Two Failure Modes

Aleatoric Uncertainty

Failure Mode 1: Data

Uncertainty in the **DATA**. The image is ambiguous (blurry, dark). It is **IRREDUCIBLE**.

Modeled via: Variance head predicting σ2.
Response: High σ2 means sensor fusion or re-capturing is needed.

Epistemic Uncertainty

Failure Mode 2: Model

Uncertainty in the **MODEL**. It hasn't seen enough of this object class. It is **REDUCIBLE**.

Modeled via: Monte Carlo Dropout (T=10).
Response: High variance means flag for human review or collect more training data.

Proposed Approach

Research Question: How does the type of uncertainty modeling affect detection reliability and calibration quality in Deformable DETR?

Pre-trained Deformable DETR (ResNet-50 • COCO • AP = 43.8)

A Track A — Aleatoric Head

WHAT

Add a parallel variance branch to the detection FFN head. Fine-tune only the new head (backbone frozen). Outputs σ^2 per detection coordination

WHY

No existing transformer-based detector outputs uncertainty — we are the first to add it to Deformable DETR.
Directly models data-level noise from Kendall & Gal.

B Track B — MC Dropout

WHAT

Keep dropout ON at test time in decoder FFN layers. Run T=10 stochastic forward passes. Variance across runs = epistemic uncertainty.

WHY

Zero retraining. Deformable DETR already has dropout in decoder. Only ~10× overhead.

C Track C — Uncertainty Signal

WHAT

Use σ^2 from Track A to compute per-image sample weights. Fine tune with weighted loss – hard images get more training attention

WHY

Uses uncertainty to improve training on hard examples. May improve APs on small/occluded objects.

Progress Till Now

Completed

- **Literature Review:** All core papers reviewed and connected to project limits.
- **Dataset Prep:** COCO val2017 (~1GB, 5k images) and 20k subset of train2017 identified.
- **Baseline Loading:** Loaded official Deformable DETR checkpoint. Verified baseline AP=43.8.

In Progress / Upcoming

- **Track A:** Variance head architecture designed; heteroscedastic loss implementation started.
- **Track B (Upcoming):** MC Dropout inference loop planned for Week 2 (T=5, 10, 20).
- **Eval Pipeline:** ECE calculation, reliability diagrams code to be built alongside Track B.

Final Deliverables

Trained Model & Eval

Deformable DETR + uncertainty heads (Tracks A/B/C) on COCO. Evaluation AP, ECE, reliability diagrams.

Ablation Study

Systematic comparison of dropout rate, T passes (5/10/20), fine-tuning epochs. Compute vs Performance analysis.

Code & Report

Full technical report, live demo with uncertainty visualization heat maps, and reproducible GitHub codebase.

Project Timeline (Mar 3 - Apr 25, 2025)

Today

Initial Presentation
Phase 1 Setup

Mar 21

Baseline Model
Implementation

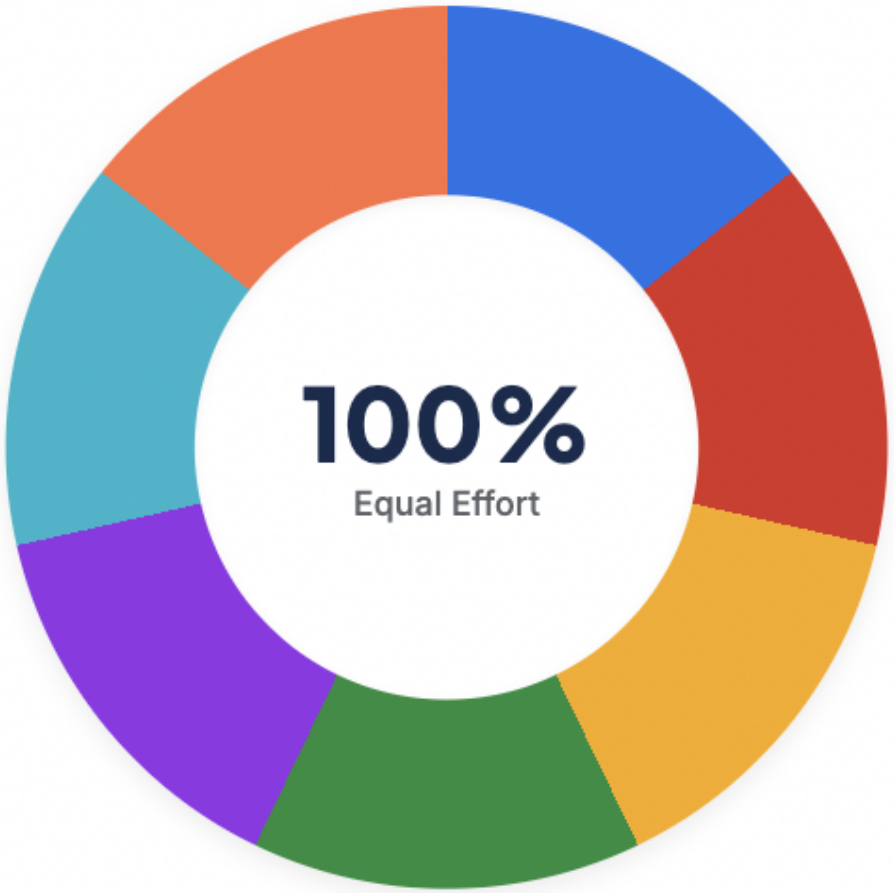
Apr 6 & 14

Results & Report
Experiments

Apr 21 - 25

Final Demo & Sub
Phase 3 Complete

Team Contributions



Our team of 7 operates with an equal distribution of workload, ensuring collaborative effort across all project phases.

Maradana Kasi Sri Roshan	14.3%
Krishna Kumar Bais	14.3%
Mahathi Garapati	14.3%
Daksh Kumar Singh	14.3%
Sontam Deekshitha	14.3%
Chinthapudi Gowtham Chand	14.3%
Sevak Baliram Shekokar	14.3%